

NSEJS

PRE-DEPARTURE

CAMP 2025

Chemistry

Structure of Atom

Lecture – 01

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Topics to be covered

1 Structure of Atom

2

3

4

5



Question



An α -particle having kinetic energy 4.0 MeV is projected towards tin nucleus ($Z = 50$). Select the correct information(s) regarding the α -particle.

$$\frac{K_{\text{initial}}}{K \cdot E} = \gamma = \frac{K Z e^2}{K \cdot E}$$

- ☒ **A** Its distance of closest approach towards the nucleus is 3.6×10^{-14} m.
- ☒ **B** Its potential energy at a distance of 9.0×10^{-14} m from the nucleus is 1.6 MeV.
- ☒ **C** Its kinetic energy at a distance of 4.5×10^{-14} m from the nucleus is 0.8 MeV.
- ☐ **D** At a moment, the distance between α -particle and the nucleus becomes 2.0×10^{-16} m.

Question



The energy of an electron in the first Bohr's orbit of H-atom is -13.6 eV. The possible energy value(s) of the excited state(s) for electrons in Bohr's orbits to hydrogen is

- ☒ A -3.4 eV
- ☐ B -10.2 eV
- ☒ C -1.51 eV
- ☒ D -0.85 eV

$$\begin{array}{cccc} & 1 & & \\ -13.6 & -3.4 & -1.51 & -0.85 \end{array}$$

(A bracket is drawn under -13.6 and -3.4)

Question



According to Bohr's atomic theory, which of the following relations is/are correct?

$$T.E = -K.E = 13.6 \frac{Z^2}{n^2}$$

$$V \propto \frac{Z}{n}$$

☒ **A** Kinetic energy of electron $\propto \frac{Z^2}{n^2}$

☐ **B** The product of speed of electron and the principal quantum number $\propto Z^2$

☒ **C** Frequency of revolution of the electron in an orbit $\propto \frac{Z^2}{n^3}$

☒ **D** Coloumbic force of attraction on the electron $\propto \frac{Z^3}{n^4}$

$$T = \frac{2\pi r}{V} = \frac{n^2 Z}{Z/n} = \frac{n^3}{Z^2}$$

$$\frac{KZe^2}{r^2} = \frac{KZe^2}{n^4/Z^2} = \frac{Z^3}{n^4}$$

Question



Some hydrogen-like atoms in ground state absorbs 'n' photons having same the energy and on de-excitement, it emits exactly 'n' photons. The energy of absorbed photon may be

no of atoms = 1

$$-13.6 \times \frac{Z^2}{n^2}$$

$$Z = 54.4 \rightarrow \text{release or absorb}$$

$$\frac{112.4}{Z=3, n=1}$$



A 91.8 eV

$$Z=2$$

B 40.8 eV

C 48.4 eV

D 54.4 eV

$$54.4$$

$$13.6$$

$$13.6$$

$$40.8$$

$$13.6$$

$$Z=3, n=2$$

$$30.6$$

$$Z=3, n=3$$

$$13.6$$

$$Z=3, n=4$$

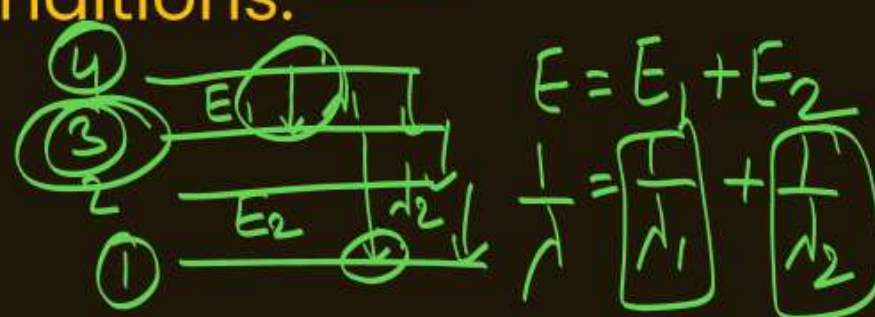
$$112.4$$

Question



The frequency of certain line of the Lyman series ($n_2 = 4$ to $n_1 = 1$) of the atomic spectrum of hydrogen can satisfy the following conditions.

$$\frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$



$$\frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$$

$$E = E_1 + E_2$$

A

It is the sum of the frequencies of a Lyman line and a Balmer line.

B

It is the sum of the frequencies of a certain Lyman line, a Balmer line and a Brackett line.

C

It is the sum of the frequencies of a Lyman line, a Balmer line and a Paschen line.

D

It is the sum of the frequencies of a Lyman and a Paschen line.

Question



Which of the following statement(s) is/are correct about the Bohr's model of hydrogen atom?

$$F = m \left(\frac{v^2}{r} \right) \quad \frac{Z^2}{n^2 \times \frac{n^2}{Z}} \propto \left(\frac{v^2}{r} \right) \quad \frac{(Z/n)^2}{n^2 Z} = \frac{Z^3}{n^4}$$

- A** The acceleration of the electron in the $n = 2$ orbit is more than that in the $n = 1$ orbit.
- B** The angular momentum of the electron in the $n = 2$ orbit is more than that in the $n = 1$ orbit.
*A.M. $\propto n h$
2H*
- C** The kinetic energy of electron in the $n = 2$ orbit is less than that in the $n = 1$ orbit.
 $\frac{1}{2} m v^2 \downarrow$
- D** The centripetal force of electron in the $n = 2$ orbit is more than that in the $n = 1$ orbit.

Question



Which of the following statement(s) is/are correct?

$$\frac{T.E}{1} = \frac{K.E}{1} = \frac{P.E}{2}$$

A The ratio of the radii of the first three Bohr orbits of hydrogen atom is 1 : 8 : 27.

B The ratio of magnitude of total energy : kinetic energy : potential energy for electron in any orbit of hydrogen atom is 1 : 1 : 2.

C The frequency of a green light is 6×10^{14} Hz, then its wavelength is 500nm.

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{500 \times 10^{-9}}$$

$$\frac{3}{5} \times 10^{15} \quad 6 \times 10^{14}$$

D The ratio of de Broglie wavelength of a H-atom, He-atom and CH₄-molecule moving with equal kinetic energy is 4 : 2 : 1.

$$\lambda = \frac{h}{\sqrt{2m(K.E)}}$$

Question



When photons of energy 4.25 eV strikes the surface of a metal 'A', the ejected photoelectrons have maximum kinetic energy T_A (in eV) and de Broglie wavelength λ_A . The maximum kinetic energy of photoelectrons liberated from another metal 'B' by photons of energy 4.20 eV is T_B ($= T_A - 1.50 \text{ eV}$). If the de Broglie wave length of these photoelectrons is λ_B ($= 2\lambda_A$), then

A the work function of 'A' is 2.25 eV .

$$\lambda_A = \frac{h}{\sqrt{2mK.E.}} \quad T_A = (K.E.) = h\nu - (h\nu_0)$$

B the work function of 'B' is 3.70 eV .

$$T_A = 4.25 - \phi_{0A} \quad \text{--- (1)}$$

C $T_A = 2.00 \text{ eV}$

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mK.E.}}$$

D $T_B = 2.75 \text{ eV}$

$$\begin{aligned} T_B &= 4.20 - \phi_{0B} \\ T_B &= (4.25 - \phi_{0A}) - 1.5 \\ T_B &= 2.75 - \phi_{0A} \end{aligned}$$

$$\lambda_A = \frac{h}{\sqrt{2mT_A}}$$

$$T_B = \frac{T_A}{4}$$

$$T_A = \frac{h^2}{2m(\lambda_A)^2}$$

$$T_B = \frac{h^2}{2m(\lambda_B)^2}$$

$$\lambda_B = 2\lambda_A$$

Question



Which of the following suggested that de Broglie wavelengths is not possible for the electron in a Bohr's orbit of the hydrogen atom?

☒ A 3.20 Å

☒ B 4.98 Å

☐ C 9.96 Å ✓

☐ D 6.64 Å ✓

$$2\pi r_n = n\lambda$$

$$2 \times 3.14 \times 0.529 = 1 \times \lambda$$

$$3.32 = \lambda_1$$

$$2 \times 3.14 \times 0.529 \times 2 = \cancel{\lambda_1}$$

$$6.64 = \lambda$$

$$r_n = 0.529 \frac{n^2}{Z-1}$$

Question



The magnitude of spin angular momentum of an electron is given by

$$s = \frac{1}{2}$$

☒ **A** $\{\sqrt{s(s+1)}\} \frac{h}{2\pi}$

☐ **B** $\frac{sh}{2\pi}$

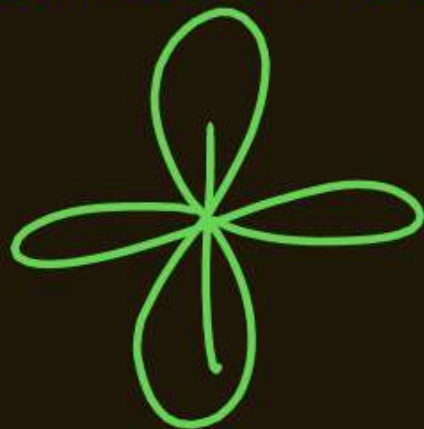
☒ **C** $\frac{\sqrt{3}}{2} \frac{h}{2\pi}$

☐ **D** $\frac{h}{4\pi}$

Question



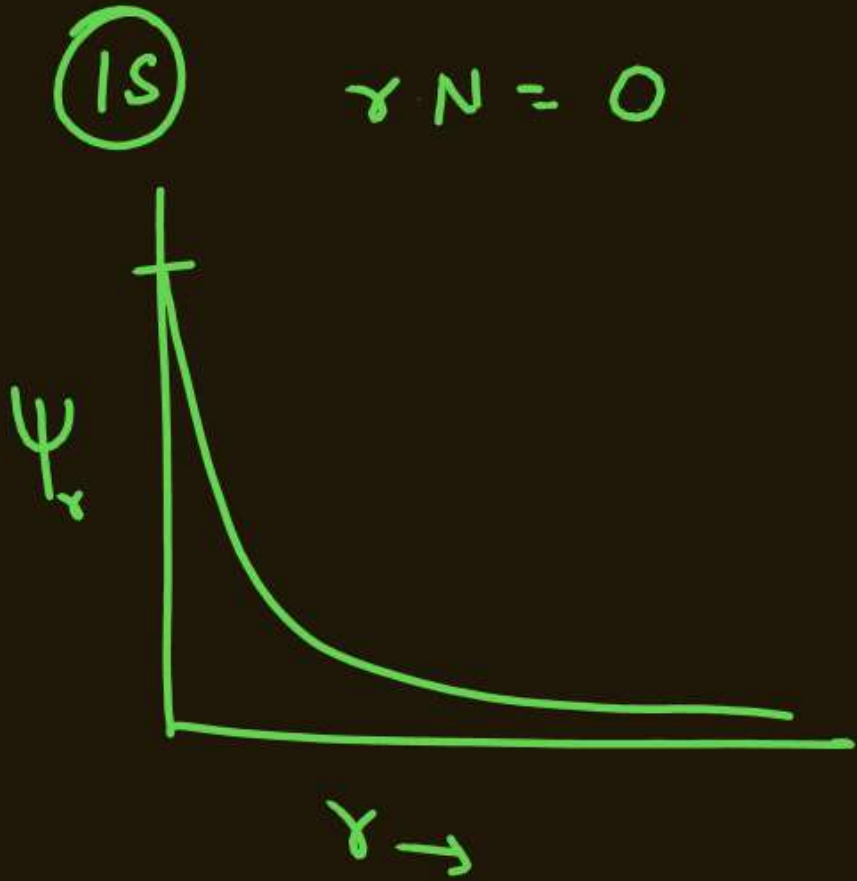
Which of the following information is true?



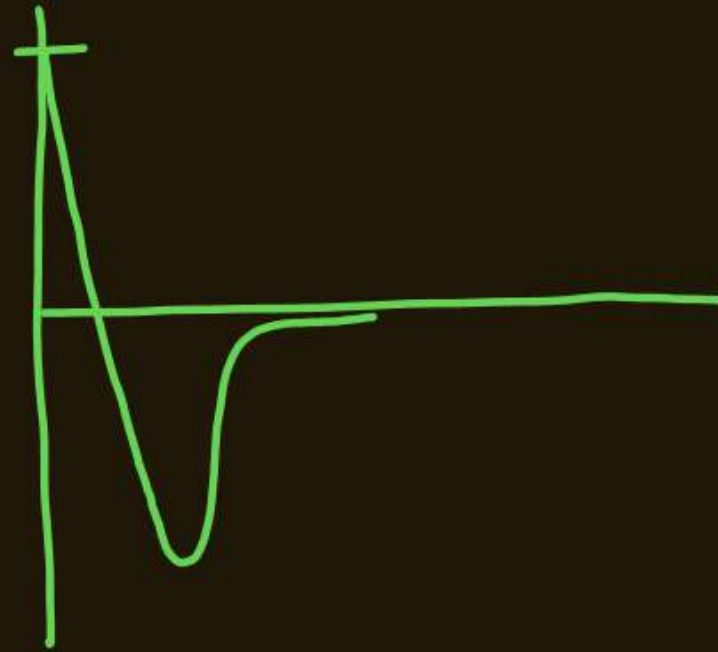
- ☒ **A** $3s$ orbital is spherically symmetrical with two nodes.
- ☒ **B** $d_{x^2-y^2}$ orbitals has lobes of electron density in XY -plane along X and Y -axis.
- ☐ **C** The radial probability curve of $1s$, $3p$ and $5d$ have one, two and three regions of maximum probability.
- ☒ **D** $3d_{z^2}$ has zero electron density in XY -plane.

$$\psi \propto r^{-1/s}$$

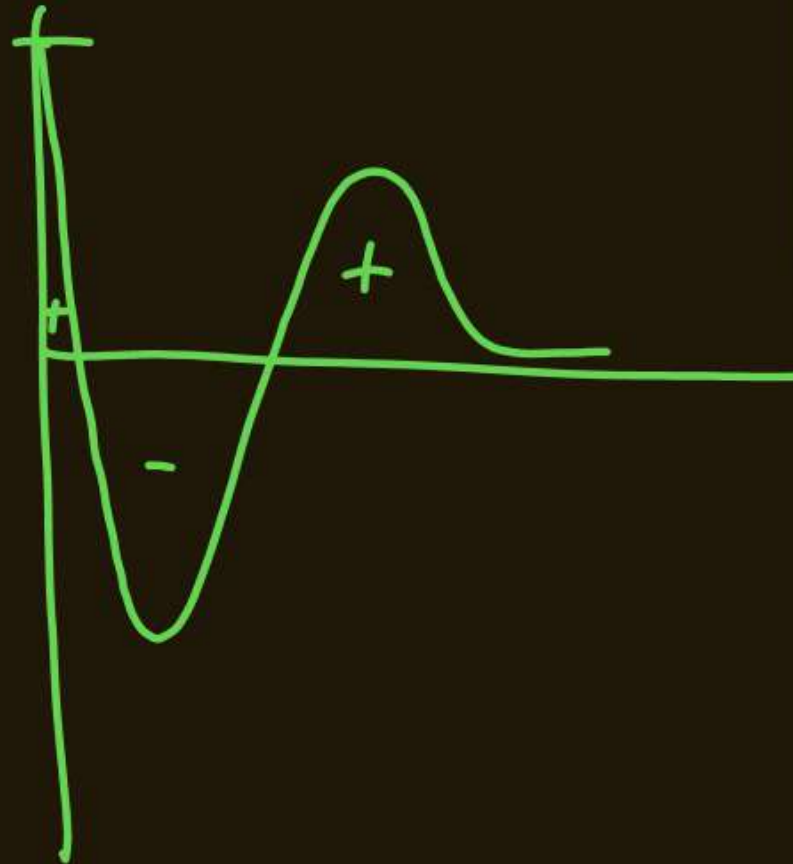
$$rN = 0$$



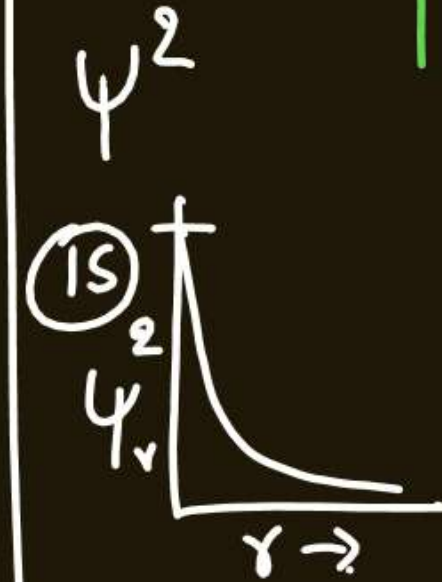
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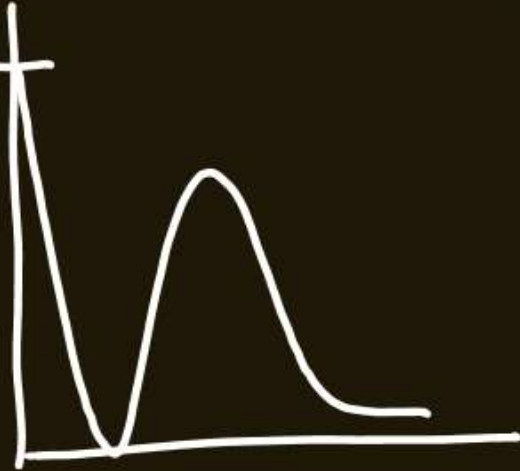
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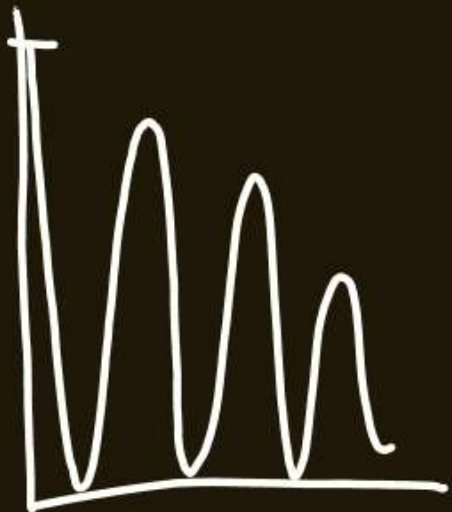
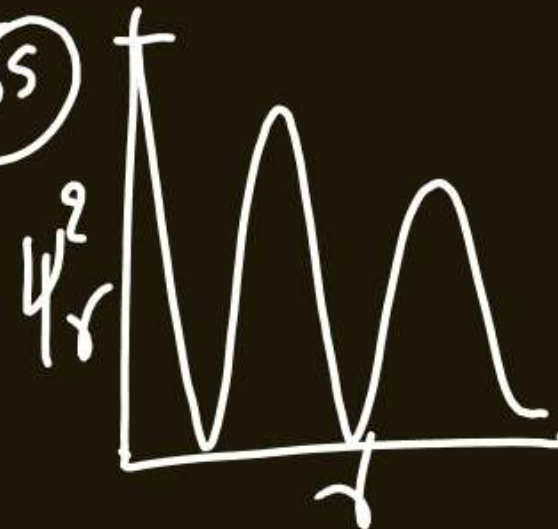
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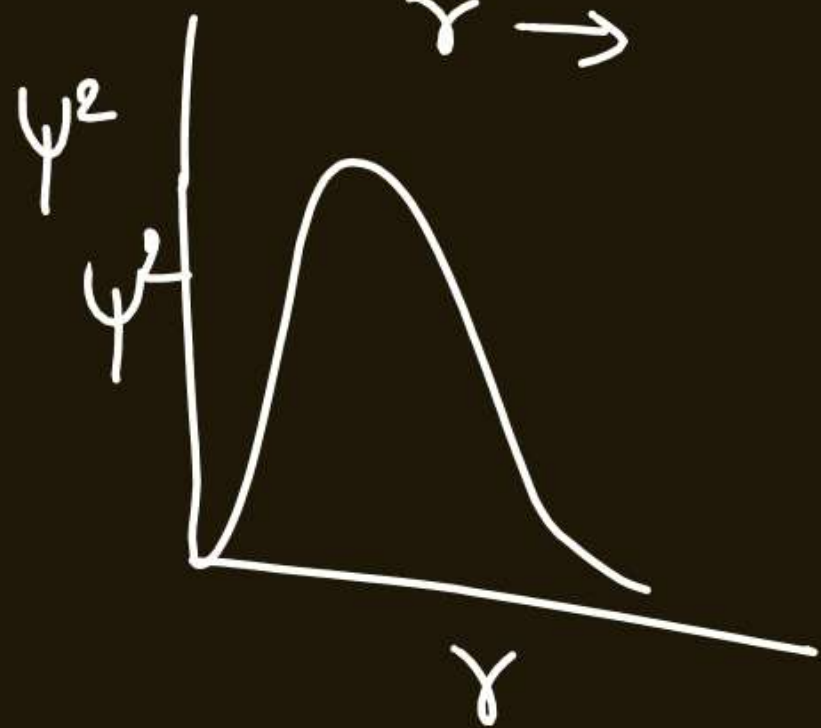
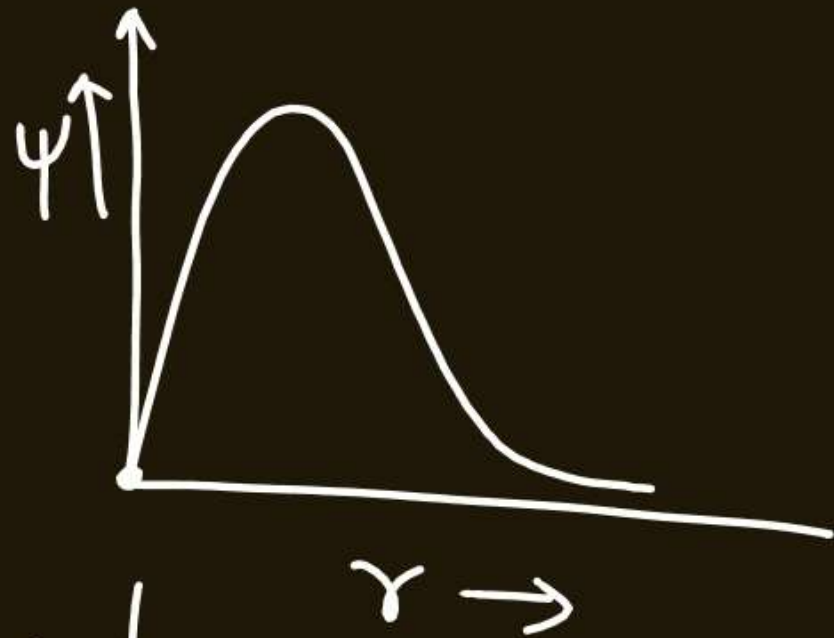
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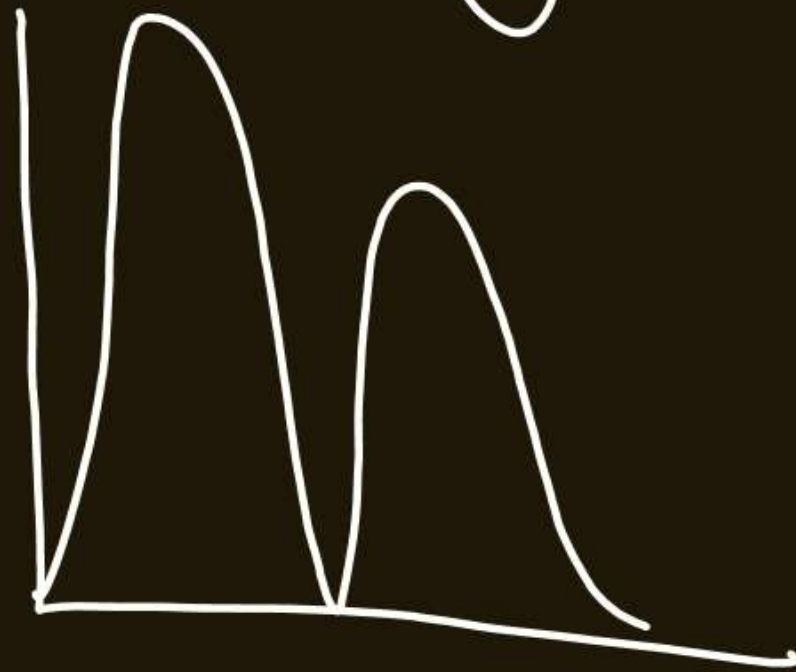
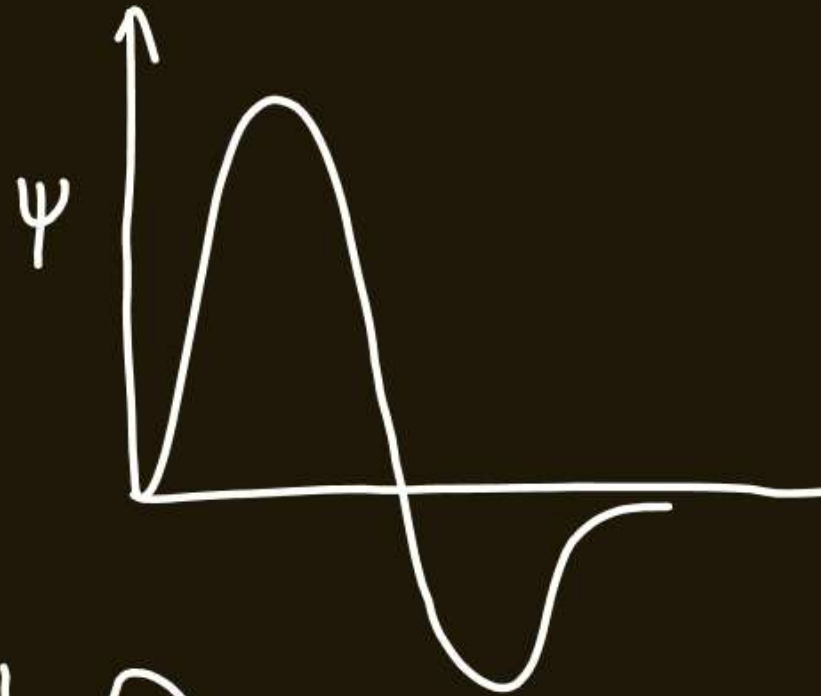
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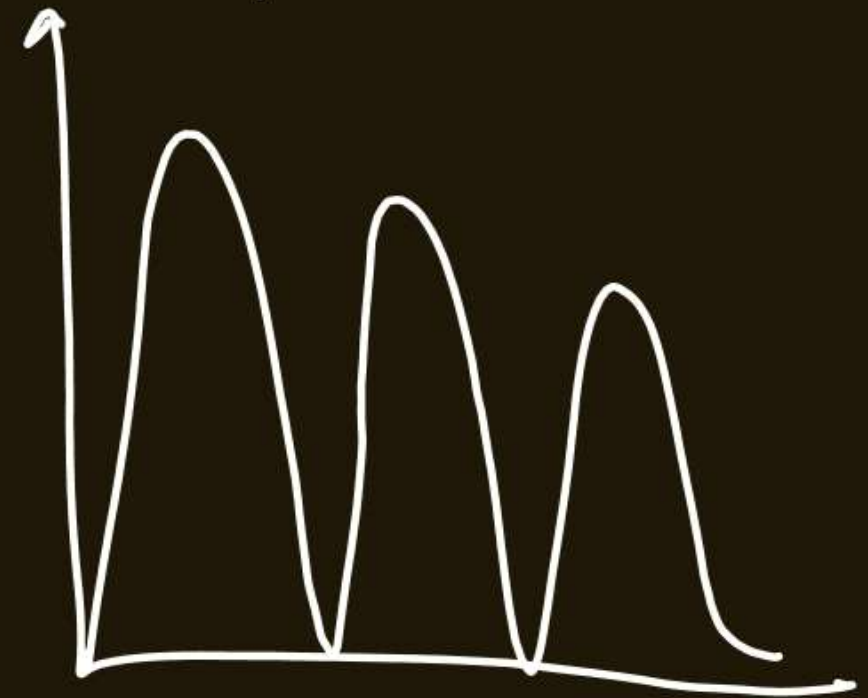
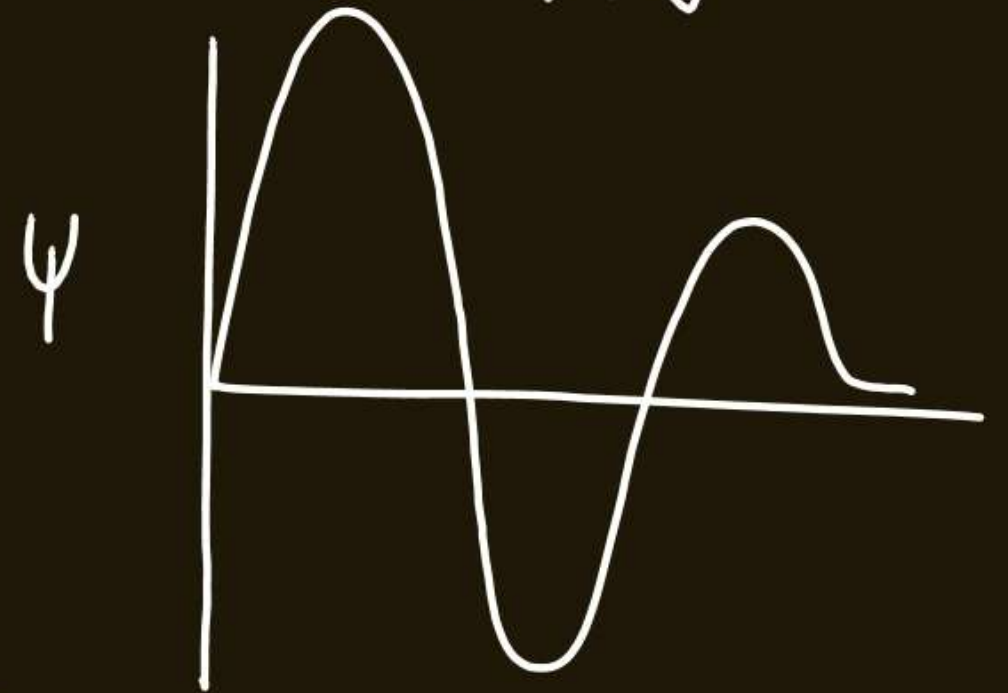
2p | 3d | 4f | 5g



3p | 4d | 5f | 6g

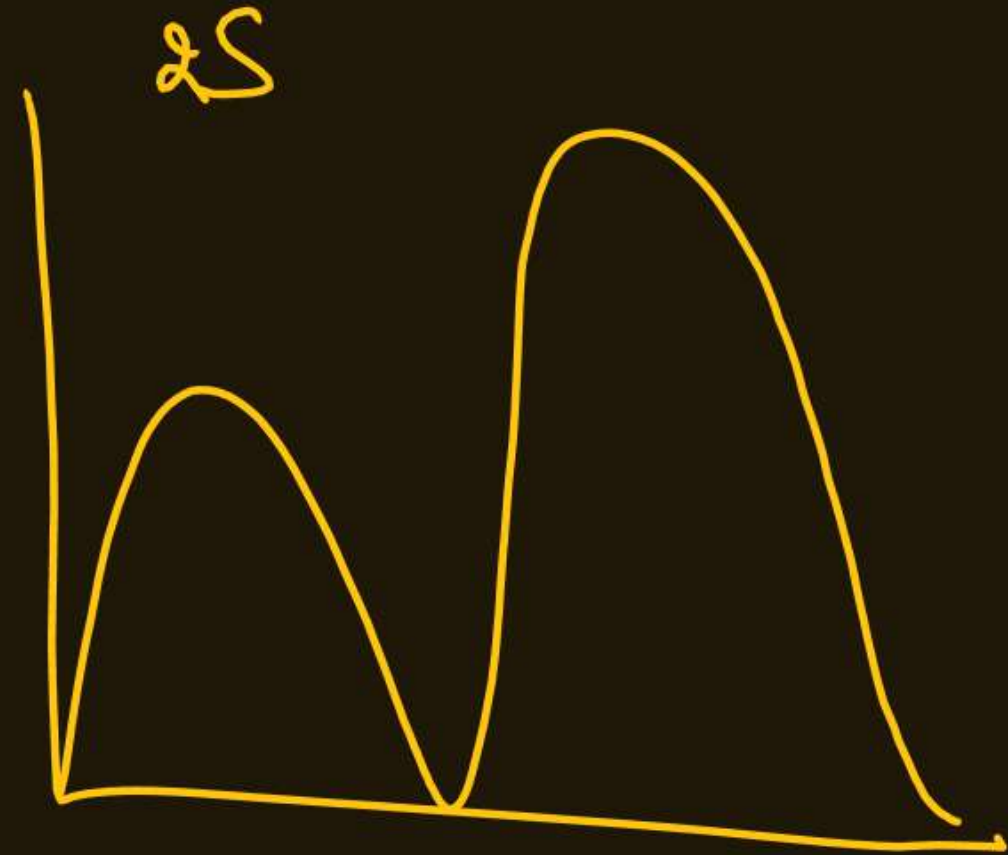
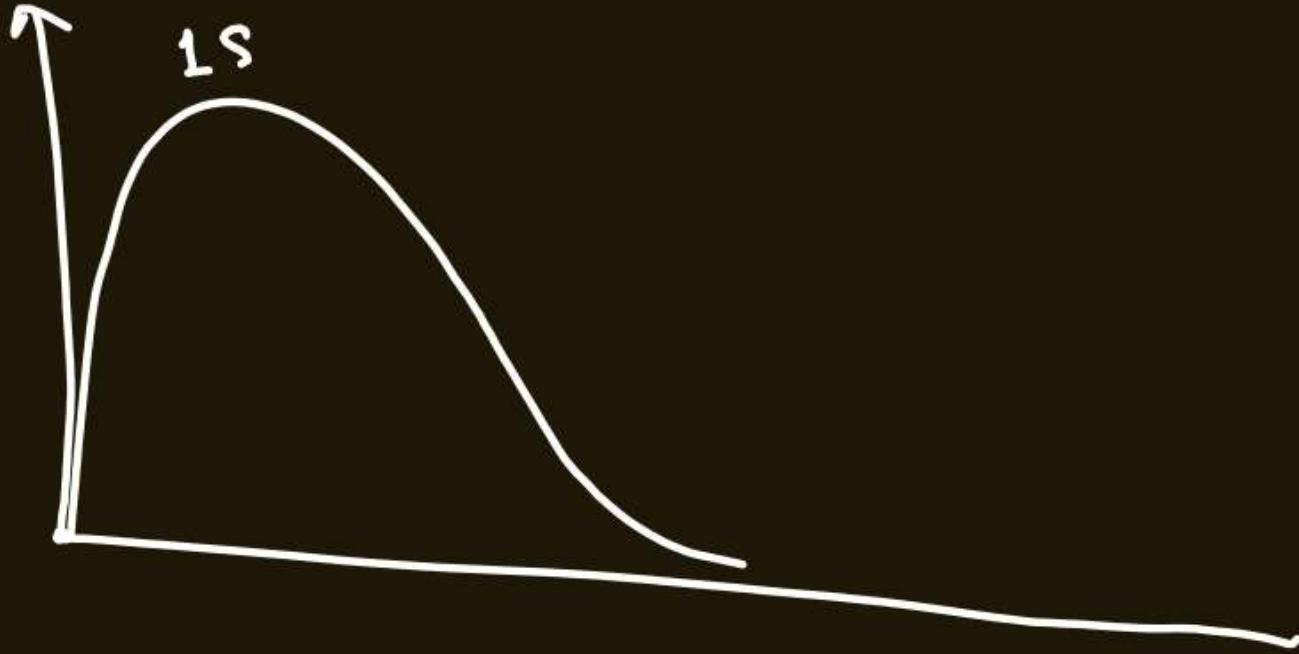


4p | 5d | 6f | 7g



$$\underbrace{4\pi r^2 \psi^2 dr}_{\text{v/s } \delta}$$

$$V \left(\psi^2 = \frac{P}{dv} \right) \times \text{Vol} = \text{Probability}$$



Question



Which is/are correct for sodium atom in the ground state?



☒ **A** There is only one unpaired electrons.

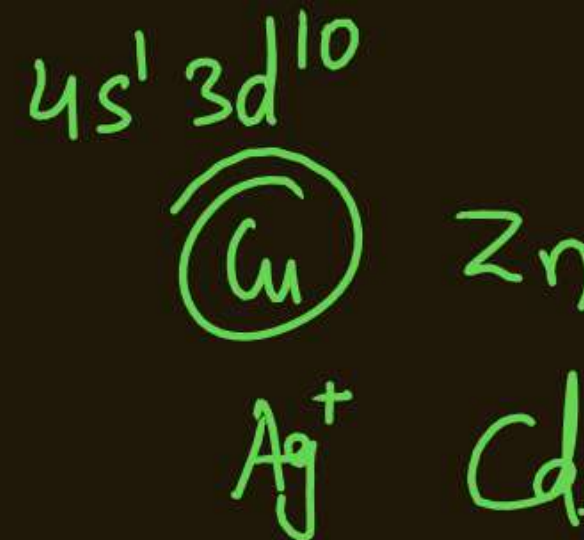
☒ **B** There are five pairs of electrons.

☒ **C** 6 electrons are in one spin and other 5 in opposite spin.

☐ **D** There are ten electrons in the same spin.

Question

Which of the following statement(s) is (are) correct?



- ☒ **A** The electronic configuration of Cr is $[\text{Ar}]3d^5 4s^1$ (Atomic number of Cr = 24).
- ☒ **B** The magnetic quantum number may have a negative value.
- ☐ **C** In silver atom, 23 electrons have a spin of one type and 24 of the opposite type.
- ☒ **D** Azimuthal quantum number may have a negative value.

THANK
YOU